meticulously removed duplicate entries and extracted preliminary information from the CSV using the versatile panda's library in Python, resulting in a refined dataset of 3,904 articles ready for further analysis. We selected the abstracts to understand the thematic pattern well and accurately identify the topics of the published papers (Principe et al., 2022).

3.2. Data preprocessing

In this work, data preprocessing consists of four main steps: 1) Text cleaning that includes removing unnecessary text and converting all text into lowercase which provides consistency and removes repetition of words. 2) Text terms such as "the", "that", "a", "in", etc., are identified as stop words without any significance are removed. This removal allows more emphasis to be granted to key terms with higher significance. 3) Tokenization involves splitting text into minimal considerable elements. It also applies bi-grams and trigrams on groups of frequent word pairs or triplets to treat them as single units, which improve the semantic analysis of specific sentences in the LDA model. 4) Lemmatization derives a word to its root form while preserving the context. For example, "organizational" will become "organization". In our work, we used the Spacy library for the above steps.

3.3. Data Processing

To better understand the research landscape within each topic, we performed an impact analysis using the litstudy (https://nlesc.github.io/litstudy/) Python library tool to extract quantitative data on the information content within our dataset. This analysis illuminated the relative influence of individual papers, authors, and journals. We calculated the impact of the paper, author, and journal based on citation count, H-index, and the impact factor of the journal (Glanzel & Moed, 2002). ¨ The litstudy library analyses textual data and calculates various impact metrics such as source of publication, year of publication, per article.

After conducting an impact analysis, we examined the content of our articles using LDA Python library Gensim (https://pypi.org/project/gensim/). LDA is capable of uncovering hidden thematic structures within the corpus. To uncover these latent themes, LDA assigns a set number of predefined topics (k) to each document iteratively, selecting the most fitting one based on the document's content. This probabilistic approach makes LDA computationally efficient, allowing us to analyze thousands of research articles effectively. We refined our corpus's textual content by eliminating punctuation, stop words (words with minimal thematic meaning), and segmenting the text into individual words (tokens). These tokens form the Document Term Matrix (DTM), a matrix that captures the frequency of each unique word within each document. We utilized Gensim's bigram and trigram functionality during tokenization to address potential ambiguities. For instance, "access" and "access control" carry distinct meanings, and this technique helps preserve these nuances within the DTM. Since LDA is unsupervised, determining the optimal number of topics (k) is crucial. We addressed this by calculating a coherence score metric. This score reflects the degree of semantic similarity between the most prominent words within each topic. Higher coherence scores indicate more interpretable and meaningful topics. Based on these scores, we selected the number of topics that produced the most semantically relevant themes. We visualized the key terms associated with each identified topic using word clouds to gain further insights. We then analyzed the most representative articles associated with each topic. This visualization provided valuable information about how blockchain technology integrates with various domains and pinpointed the most recent trending research areas in blockchain adoption.

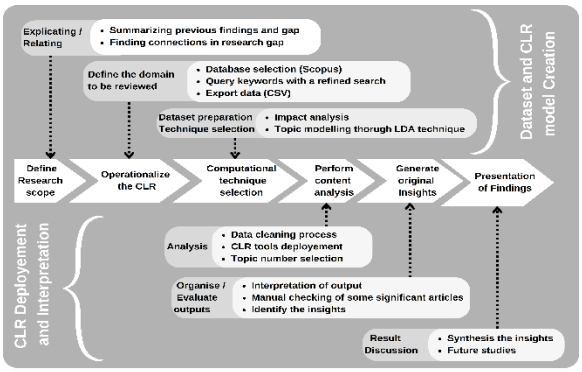

Figure 1. Research framework

In our research, we used the CLR method on the relevant literature, as depicted in Figure 1. The interpretation of LDA output and in-depth analysis within each topic follows an iterative cycle for optimal results. This process involves refining the analysis by adjusting parameters or diving deeper into specific topic clusters to ensure accurate and impactful conclusions**.**

4. Results

4.1. Impact analysis

We conducted an impact analysis on various research areas and topics based on factors such as the

citation of articles, authors, and journals generated by the research community. After analyzing the data, we discovered that most of the recently published articles had little or no citations, mainly because the keywords we selected for this study were not very popular before 2016, which resulted in fewer papers to analyze. Therefore, we decided to begin our impact analysis from 2016. We have presented the number of papers published each year and how many times they were cited in Figure 2.

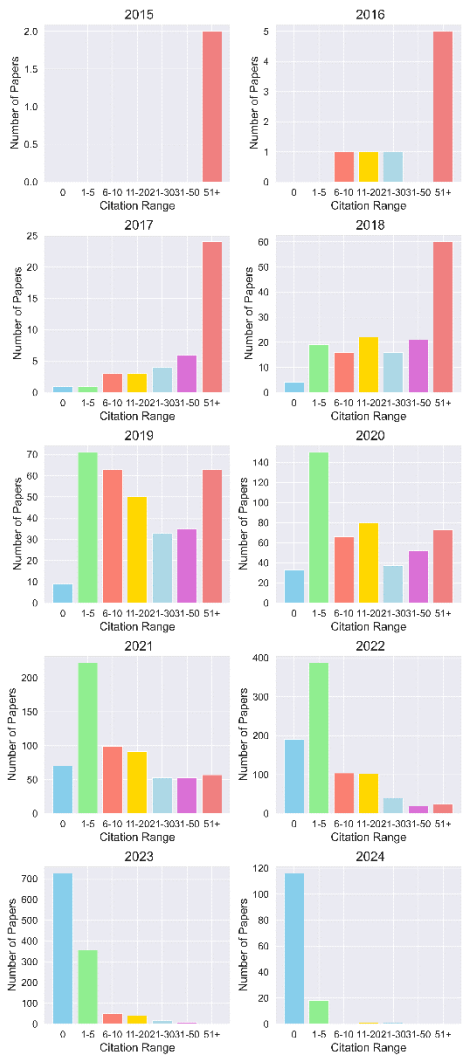

Figure 2. Number of publications vs Citation/year

In Figure 3, we individually present the number of publications per year to clarify the analysis of publications and citations. This visualization demonstrates the recent impacts of blockchain for information security and privacy research growth in academia. The growing interest started in 2017 and continued till 2020, when more than 1200 articles were published that predict the future research trends related to topics highlighted in different domains. In addition, we have picked out the twenty best journals and conferences based on their high number of published papers, as shown in Figure 4.

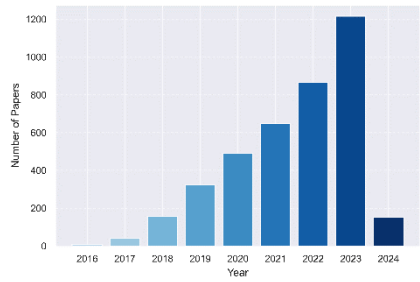

Figure 3. Publication Per Year

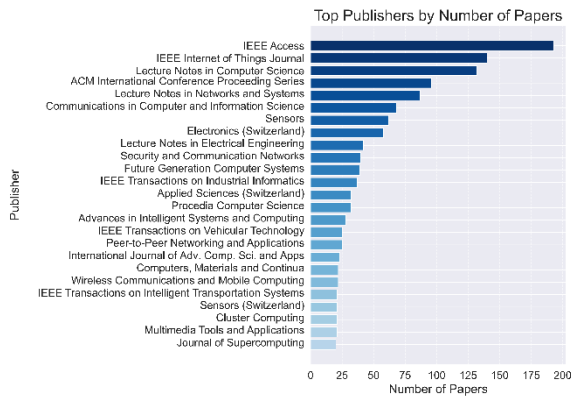

Figure 4. Top Publishers

In Figure 4, we found included journals from Information Systems, such as the IEEE Internet of Things Journal, IEEE Transactions on Industrial Informatics, and IEEE Transactions on Intelligent Transportation Systems. Including these IS journals Indicates That the IS encourages the advancement of security and privacy using blockchain. Also, these journals are essential in our research as they are related to developing secure and advanced infrastructure using IoT technology with the help of blockchain (Lu et al., 2019). These journals and conferences influence the impact of the selected topics

4.1. Content analysis

In our content analysis, we determined the number of topics and verified their accuracy by analyzing the coherence score. Figure 5 displays the coherence score of the topics from our topic modeling process. After reviewing the results in Figure 5, we chose 30 topics

with a coherence score higher than 0.4 as these topics indicate high semantic relation in the corpus. Previous study also indicates that similar coherence scores produce better topic modeling outcomes (Stevens et al., 2012). These topics are visualized in a word cloud in Figure 6.

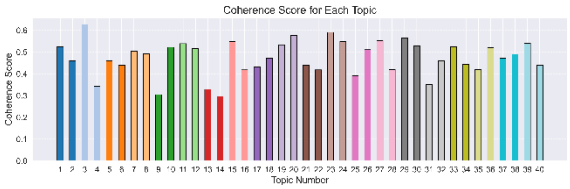

Figure 5. Coherence Score of topics

Figure 6. Word cloud of topics

We analyzed the word cloud and merged related topics to develop 10 new topics based on research areas for security and privacy in blockchain technology. These topics span different domains and are summarized as 1) Privacy of personal information in Healthcare, 2) Identity Management in E-education and E-voting systems, 3) Security Measures using IDS, 4) Data privacy on Cloud and Edge computing, 5) Access and authentication management in supply chain using smart contracts, 6) Advanced encryption protocols using ZKP, 7) Privacy protection in Distributed Systems using FL, 8) Security and privacy in IoT-driven systems and smart cities, 9) Data privacy in Energy trading over the smart grid, 10) Privacy of real identity in Metaverse. We have investigated to understand the evolution of topics from 2016 to 2024. In Figure 7, the topic trend indicates that the focus on blockchain-based security and privacy has increased in recent years, specifically for the supply chain, healthcare and IoT-driven systems because of the rising concern towards information security and privacy. The pattern shows the high interest of academia and industry in performing research on secure data and privacy protection in the blockchain for digital transformation in growing fields like computing, IoT, cloud, and federated learning. The surge in

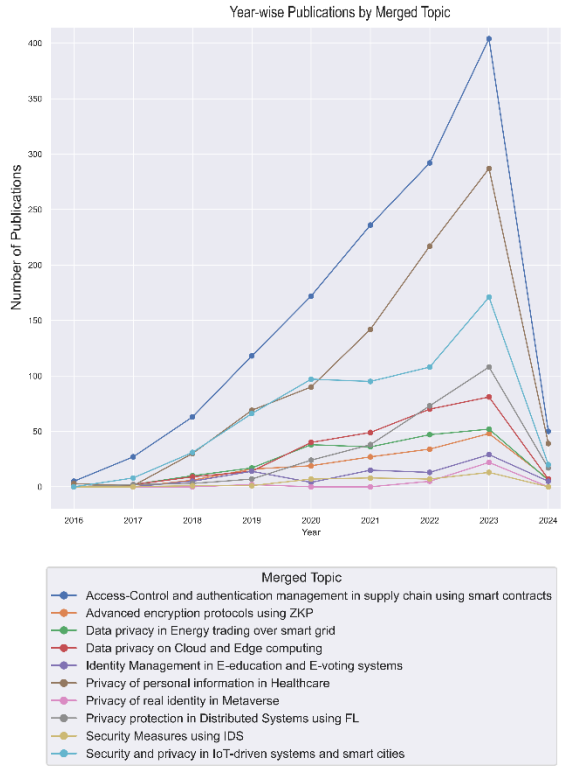

Figure 7. The evolving topics trend over the year

publications after 2020 addresses the critical role of emerging technological threats. This topic trend analysis indicates that ongoing innovation and adaptive security measures will be essential to future research directions

5. Discussion

The CLR finding provides impact and content analysis for blockchain-based information security and privacy, which helps to use resources in right direction in multidisciplinary research for Industry and academia. To clarify this, Figure 8 represents the identified topics with related keywords. We selected most impactful articles on each topic through detailed content analysis and offered an in-depth discussion with relevant sources.

T1) Privacy of personal information in healthcare: Researchers are focusing on enhancing privacy, security, and trust mechanisms through smart contracts and ZKP for self-sovereign identities. However, challenges like high implementation costs, legal issues, and complexity remain barriers to adoption (Liu et al., 2020).

T2) Identity Management in E-education and E-voting systems: Utilization of advanced cryptographic techniques in a smart contract ensures secure